The web server puts whatever path you visit, such as /String/each, into the PATH_INFO environment variable. We break this text into the String class and the each prefix, get a list of the instance methods that belong to String, and finally narrow them down to the ones that start with each.

We've already applied the lessons from earlier in this chapter and injected the output collaborator via constructor injection. It might be tempting to pass in a spy from our tests so that we could check that the CGI script was writing the correct results:

15-using-test-doubles-effectively/19/ruby_doc_server/spec/ruby_doc_server_spec.rb

```
require 'ruby_doc_server'

RSpec.describe RubyDocServer do
  it 'finds matching ruby methods' do
    out = get('/Array/max')

    expect(out).to have_received(:puts).with('Content-Type: application/json')
    expect(out).to have_received(:puts).with('["max","max_by"]')
  end

  def get(path)
    output = object_spy($stdout)
    RubyDocServer.new(output: output).process_request(path)
    output
  end
end
```

Unfortunately, this spec is quite brittle. It's coupled not just to the content of the web response, but also to exactly *how* it gets written.

Ruby's IO interface is large. It provides several methods just for writing output: puts, print, write, and more. If we refactor our implementation to call write, or even to call puts just once with the entire response, our specs will break. Recall that one of the goals of TDD is to support refactoring. Instead, these specs will stand in our way.

Test Doubles Are Best for Small, Simple, Stable Interfaces



Large interfaces aren't the only ones that are hard to replace with a double. We also see problems in the following cases:

- *Complex interfaces:* The more complex the interface, the harder it is to mimic accurately; we like to use test doubles to steer our design toward simplicity.
- *Unstable interfaces:* Every message you expect or allow is a detail your specs are coupled to; the more the interface changes, the more you're going to have to update your doubles.